Topic 2: Data

Computers use binary to represent different types of data.

Students are expected to learn how different types of data are represented in a computer.

Subject content	Students should:	
2.1 Binary	2.1.1	understand that computers use binary to represent data (numbers, text, sound, graphics) and program instructions and be able to determine the maximum number of states that can be represented by a binary pattern of a given length
	2.1.2	understand how computers represent and manipulate unsigned integers and two's complement signed integers
	2.1.3	be able to convert between denary and 8-bit binary numbers (0 to 255, -128 to +127)
	2.1.4	be able to add together two positive binary patterns and apply logical and arithmetic binary shifts
	2.1.5	understand the concept of overflow in relation to the number of bits available to store a value
	2.1.6	understand why hexadecimal notation is used and be able to convert between hexadecimal and binary